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Techno-economic analysis of sustainable biofuel production from whole olive waste

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ABSTRACT

Olive waste has a high potential for biofuel production, while it is typically buried or incinerated, contributing to significant environmental issues. This study presents a technical and economic analysis of bioethanol, biogas, and pure lignin production from various olive wastes under different scenarios. The most promising methods for lignin removal are organosolv pretreatment and acid-catalyzed organosolv pretreatment, which are evaluated under two sets of scenarios A and B, respectively. These scenarios include sub-scenarios of A-1 and B-1 (olive stone), A-2 and B-2 (olive leaves), A-3 and B-3 (olive pomace), A-4 and B-4 (olive wood), and A-5 and B-5 (mixture of all wastes). Following pretreatment, ethanol was produced via presaccharification and simultaneous saccharification and fermentation, while the remaining wastewater was converted into biogas. Process simulations were performed in Aspen Plus, and economic evaluations were conducted by Aspen Process Economic Analyzer. The highest gasoline equivalent prices were obtained for scenarios A-1 (1.03 $/L) and B-1 (0.87 $/L), whereas the lowest values were observed in scenarios A-5 (0.56 $/L) and B-5 (0.57 $/L). Furthermore, the highest capital expenditures and operational expenditures were observed in scenarios B-5 (77 and 25.62 M$) and A-5 (75.75 and 25.6 M$), while the lowest costs corresponded to scenarios B-1 (31.96 and 3.40 M$) and A-1 (33.4 and 3.7 M$), respectively. The optimal case was 160,000 tons/year of olive waste mixture with payback periods of about 4.3 years for scenarios A-5 and B-5. These results demonstrate that a biorefinery producing bioethanol and biogas from olive waste is economically viable.

1. Introduction

Greenhouse gas (GHG) emissions derived from fossil fuel consumption have instigated adverse consequences such as global warming, drought conditions, and erratic climate alterations [1,2]. According to a World Bank report, global carbon dioxide emissions increased by 300 % between 1960 and 2014, with Iran ranking seventh globally, contributing 1.8 % of the total CO_2 emissions [3]. This environmental crisis has compelled global communities to seek clean and renewable alternative energy sources that are not only environmentally friendly but also economically cost-effective [4]. In this context, biomass, as a renewable and sustainable resource, is considered a promising alternative due to its potential for producing high energy density liquid biofuels [5].

Statistics indicate that in 2019, approximately 11 % of the world's total energy and 2.13 % of Iran's total energy were supplied through renewable energy technologies [6]. Given the limitations of first-generation biofuels (derived from food-based sources such as wheat) and the lack of economic feasibility in the production of third-generation (algae-based) and fourth-generation (engineered biomass-based) biofuels [7]. Second-generation biofuels- produced from

Abbreviations: GHG, Greenhouse gas; OTP, Olive tree prunings; FAO, Food and Agriculture Organization; NREL, National Renewable Energy Laboratory; CO_2, Carbon dioxide; CH_4, Methane; UASB, Upflow Anaerobic Sludge Blanket; COD, Chemical oxygen demand; AD, Anaerobic digestion; NRTL, Non-random two-liquid; SRK, Soave-Redlich-Kwong; APEA, Aspen Process Economic Analyzer; SSF, Simultaneous saccharification and fermentation; PSSF, Presaccharification and Simultaneous Saccharification and Fermentation; IRR, Internal rate of return; PP, Payout Period; PI, Profitability Index; n, project timelife; i, discount rate; CF_t, net cash flow in year t; CAPEX, Capital expenditure; OPEX, Operational expenditures; PSC, Product selling costs; PEC, Purchased Equipment Cost; HP, High-Pressure steam; LP, Low-Pressure steam.

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lignocellulosic materials that are abundant, carbohydrate-rich, and non-competitive with food production- are expected to gain significant importance in the near future [8,9].

Among liquid biofuels, bioethanol is commonly mixed with gasoline in different proportions to improve combustion efficiency [10]. The advantages of using bioethanol include reduced greenhouse gas emissions compared to fossil fuels, a high octane number, and a significant fuel heating value [11]. In recent years, researchers have technically and economically evaluated a variety of plant and biological waste resources for the production of biofuels, including bioethanol, biogas, and biodiesel. These resources include wheat straw [12], municipal solid waste [13], citrus peel waste [14], palm oil waste [15], sugarcane bagasse and brown algae [16], sweet sorghum [17], textile waste [11], woody biomass [18], olive tree prunings (OTP) [19], olive stone [20] and olive pomace [21].

The olive industry is important in the world. According to the Food and Agriculture Organization (FAO), Iran is among the olive-producing countries, with around 112,000 tons of olives produced in 2022 [22]. This industry's by-products, such as olive tree pruning residues, olive stone, and different types of olive pomace, have high quantities of cellulose, hemicellulose, and lignin [23]. Compositional analysis shows that olive pomace possesses 20–30 % of carbohydrates while olive stones contain 28.1–40.4 % of cellulose, 18.5–32.2 % of hemicellulose, and 25.3–27.2 % of lignin. Further, biomass from olive tree pruning has a high carbohydrate content ranging from 46.1 % to 61.6 % [24]. Thus, the abundant carbohydrate content in olive-derived biomass suggests its potential as a feedstock for biofuel and other fermentation product production.

Notably, the bioethanol production process from olive waste generates residues whose disposal may raise environmental concerns. Nevertheless, due to their high carbon content and rich nutrient composition, these agricultural residues are considered a suitable feedstock for anaerobic digestion [25]. Anaerobic digestion has been introduced as an attractive technique for achieving clean and sustainable energy, as well as an efficient and environmentally friendly method for waste management [26,27].

In this process, microorganisms decompose the waste materials, converting them into energy forms such as biogas and biofertilizers [28]. For the economical production of biofuels, various pretreatment methods, including steam explosion [29], dilute sulfuric acid [30], NaOH [31], acetone-sulfuric acid [32], and acid-catalyzed organosolv [33] have been evaluated to prepare different substrates and enhance fermentation efficiency.

One of the main obstacles to using lignocellulosic materials for fermentation and ethanol production processes is lignin's complicated molecular structure [34]. One of the most effective techniques for lignin removal is organosolv pretreatment [35]. In this pretreatment, lignin separation before enzymatic hydrolysis produces a useful byproduct and also lowers costs and improves process efficiency [36]. Considering the inhibitory effects of organic solvents on the enzymatic hydrolysis and fermentation stages, it is essential to remove them as much as possible before fermentation. Solvent recovery is also necessary to reduce costs and environmental impacts [36]. Acid-catalyzed organosolv pretreatment is used for enhanced lignin separation [37].

Studies have shown that the cost of raw materials [38], the pretreatment method [21], the production of valuable by-products [19], and the plant capacity [39] are among the key parameters determining the economic viability of biofuel production processes. Numerous studies have investigated the potential of olive tree pruning residues, olive pomace, and olive stone for biofuel production. To our knowledge, a comprehensive assessment of the potential of all waste materials and by-products derived from olive cultivation and industry (as an integrated biorefinery for the production of bioethanol, biogas, and other value-added products) has not yet been conducted. The objective of this project is a technical and economic feasibility study of bioethanol production as the main product, along with biogas, lignin, and biofertilizer as by-products, from all waste materials generated by olive cultivation and industry. The goal is to maximize the reuse of these residues while minimizing environmental impacts and improving waste management practices, ultimately moving towards a circular economy. Based on experimental data [40], the simulations of the main equipment for two sets of scenarios using all olive residues, both individually and as mixtures, with organosolv and acid-catalyzed organosolv pretreatment, were performed using Aspen Plus software v11. Subsequently, based on the gained simulation results, equipment sizing and optimization were conducted using Aspen Process Economic Analyzer (APEA). Finally, a sensitivity analysis of the process was performed to evaluate the impact of raw material costs and by-product prices on economic indicators.

2. Methods

To produce bioethanol and biogas from olive waste, a study was conducted by designing ten scenarios. These scenarios (A and B) comprised two five-scenario groups: the first group (scenario A) aimed at producing bioethanol and biogas using organosolv pretreatment, and the second group (scenario B) utilized acid-catalyzed organosolv pretreatment. In this regard, the process simulation was performed for all olive waste, both individually and as a mixture, with two types of pretreatments. In all scenarios, presaccharification and simultaneous saccharification and fermentation (PSSF) was employed for biofuel production [41]. The ethanol obtained was purified to a fuel-grade purity of 99.9 % (w/w). Additionally, during the distillation and dehydration stages, a CO_2 stream with over 98 % w/w purity was extracted, possessing the potential for use in dry ice production, beverage industries, and food storage [11,42].

2.1. Set of A scenarios

2.1.1. Feed Handling

In this study, raw materials (on a dry weight basis) are continuously supplied to the plant, including 15 ktons/year of olive stone (scenario A-1), 40 ktons/year of olive leaves (scenario A-2), 40 ktons/year of olive pomace (scenario A-3), 65, ktons/year of olive wood (Scenario A-4), and 160 ktons/year of a mixture of all olive wastes (scenario A-5). Scenarios A and B, including their sub-scenarios, are depicted in Table 1.

For feedstock preparation, olive tree prunings (including leaves and wood) are collected and dried in the field, then transported to the factory along with olive stones. Olive mill leaves are produced in the oil mill during the washing/cleaning stage, while olive tree pruning garbage is obtained in the field post fruit gathering and is made up of leaves, small twigs, and timber [24]. Since leaves are collected primarily in autumn and wood until the end of winter, they were therefore examined separately [43]. The olive pomace produced during the oil extraction process is also shredded using a grinder after drying and stored in bulk form. Also, only the wastewater generated during the refining process has been studied and modeled. As shown in Fig. 1, after feedstock preparation, the materials are transferred to the pretreatment unit via a

Table 1
Definition of Scenario A and B and their sub-scenarios.

Set of Scenarios	Raw material	Pretreatment	Scenarios
Set of A Scenarios	Olive stone	Organosolv	A-1
	Olive leaf	Organosolv	A-2
	Olive pomace	Organosolv	A-3
	Olive wood	Organosolv	A-4
	Mix	Organosolv	A-5
Set of B Scenarios	Olive stone	Acid-catalyzed organosolv	B-1
	Olive leaf	Acid-catalyzed organosolv	B-2
	Olive pomace	Acid-catalyzed organosolv	B-3
	Olive wood	Acid-catalyzed organosolv	B-4
	Mix	Acid-catalyzed organosolv	B-5

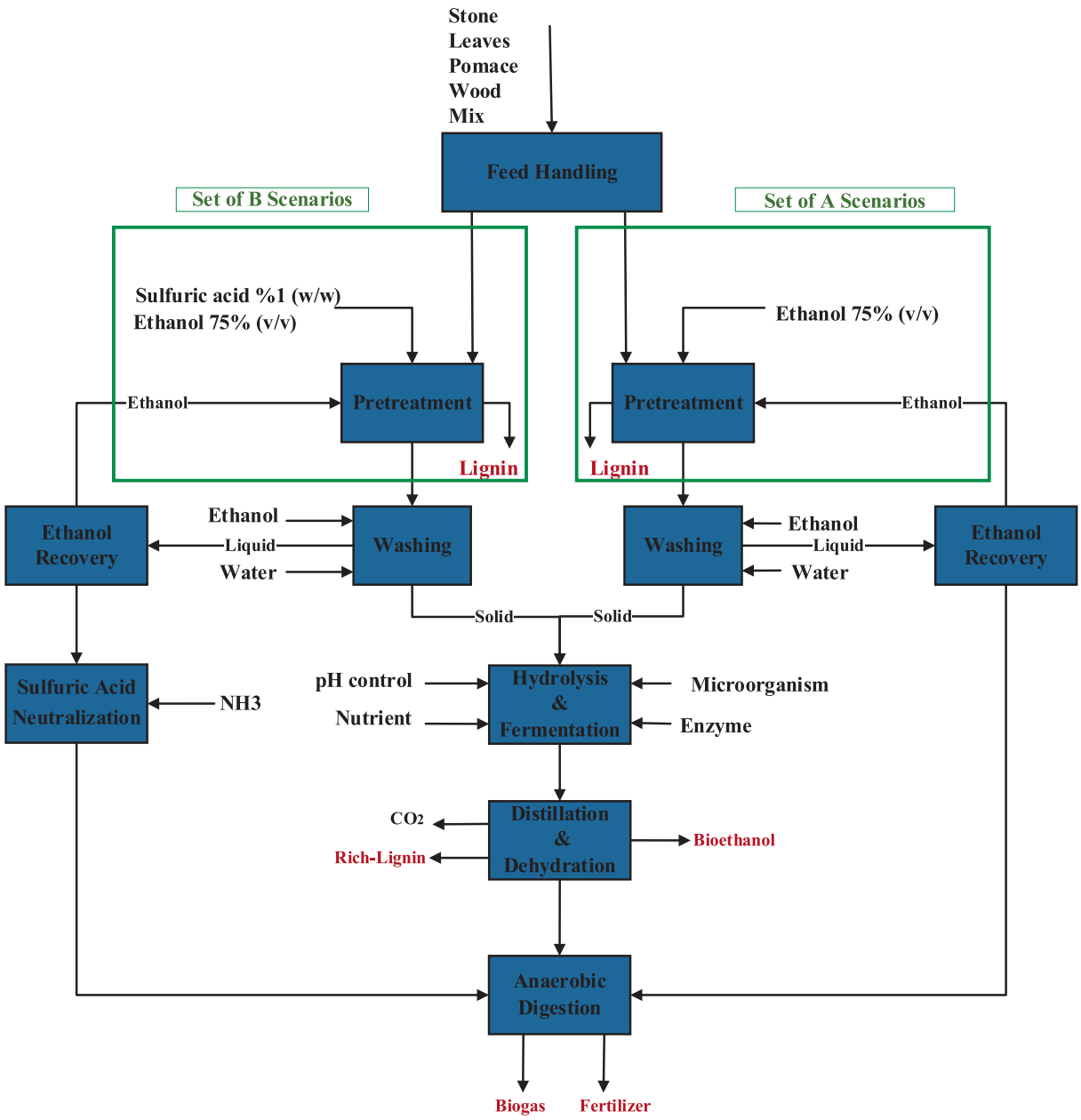


Fig. 1. Block flow diagram of olive waste-based biorefineries for scenario set A (right; organosolv pretreatment) and scenario set B (left; acid-catalyzed organosolv pretreatment).

conveyor belt.

The main units considered include various stages such as feedstock preparation, pretreatment, hydrolysis and fermentation, ethanol production and purification, lignin separation, ethanol recovery, sulfuric acid neutralization, and biogas production and upgrading. Fig. 1 depicts the block flow diagram (BFD) of the process for both sets of A and B scenarios. The differences in the structural composition of olive tree pruning residues, olive pomace, and olive stones, as well as the varying capacities of each substrate and the effectiveness of different pretreatment methods, lead to distinctions in performance and outcomes among the various scenarios studied.

The weighted average mass fraction for each type of waste is used. In other words, the composition of the mixed feed is the sum of the composition of each component multiplied by the mass fraction of that component in the mixture. The percentage composition of the components in olive waste is presented in Table 2.

Table 2
The composition of different parts of olive waste plants (based on w/w%) [40].

Component	Stone	Leaves	Pomace	Wood	Mixture of all waste
Glucan	38.27	53	38.08	33.23	38.97
Galactan	9.84	20.64	14.42	30.99	22.08
Xylan	23.36	8.34	19.04	16	15.53
Lignin	28.53	18.02	28.46	19.78	22.42

2.1.2. *Pretreatment*

In the A scenario set, all olive waste (both mixed and separated) is introduced into the pretreatment reactor after preheating. All pretreatment processes were carried out at a solid-to-liquid ratio of 1:8, at 180 °C for a period of 1 h. In this scenario, 75 % ethanol was used for the pretreatment [40]. By changing the structure of lignocellulosic materials, this process facilitates the cellulase enzyme's access to cellulose and hemicellulose. In this simulation, an ethanol–water solution of 75 %

(v/v) is used to treat the feedstock as it enters the pretreatment unit. The pretreatment reactor outlet, a flash two-phase separator, removes some of the formed volatile inhibitors. After the flash step, a filtration process separates the liquid and solid fractions [33]. Afterward, the pretreated solids are first fed into a separation filter and then directed to two four-stage counter-current solid-washing units, where a solution of 75 % ethanol and water is used to remove residual sugars and ethanol from the pretreated biomass. The intensity of washing and the volume of water must be sufficient to prevent inhibitors from entering the hydrolysis and fermentation units [32], thereby avoiding toxicity. The liquid phase is initially transferred to a settling tank for lignin separation and then to a distillation column for ethanol recovery.

2.1.3. Hydrolysis and fermentation

In simultaneous saccharification and fermentation (SSF), the enzymatic reaction operates at a temperature significantly lower than the optimal temperature for the enzyme, which reduces the enzyme's activity potential and consequently increases enzyme demand. Therefore, a PSSF process was used [41]. In this process, the cellulase enzyme known as Cellic® CTec2 (Novozymes, Denmark) was first added to the substrate at an optimal temperature of 45 °C. Following 24 h of enzymatic hydrolysis, the yeast Saccharomyces cerevisiae (CCUG 53310, Culture Collection, University of Gothenburg, Sweden) was added to the reactor after cooling it to 37 °C [44]. The residence time for fermentation and conversion of sugar to products is 72 h.

2.1.4. Ethanol distillation and dehydration

For ethanol purification, a distillation and dehydration unit was employed, consisting of a beer column, a rectifier, and a scrubber column. The beer column with 22 sieve trays was filled with the fermentation product, which was preheated before being transferred. This column operates at a vacuum pressure of 0.97 bar, with a tray efficiency of 80 %. In the beer column, the feed purity increases from 10 % to 57 % ethanol by weight. The ethanol vapor produced in the beer column enters tray 15 of the stripper column. These vapors serve as a heating medium for the second distillation column (rectifier) [45]. This column has 30 sieve trays (with 80 % efficiency) and operates at a pressure of 0.9 bar [11]. In the second column, ethanol with a purity of 93 % by weight is produced. The bottom product of the rectifier column, which is nearly pure water, has a purity of 98 % by weight and is processed for the production of clean water. The ethanol was further purified to produce ethanol with 99.9 % purity by weight by passing the overhead product from the second column to a molecular sieve unit. The CO_2 generated from the fermentation process and the non-condensable gases (containing ethanol) exiting the beer column were fed into a seven-stage scrubber (packed) with an 80 % performance efficiency. The gas flow entered from the bottom, while the washing water was fed from the top of the scrubber column. About 97 % of the CO_2 entering the distillation unit exited from the top of the scrubber, and the bottom flow of the scrubber was returned to the beer column. The bottom product from the beer column was transferred to a separator to separate lignin and other solids. The outlet stream from the bottom of the separator, which is lignin-rich, was stored in a tank.

2.1.5. Lignin separation

The outlet stream from the bottom of the separator is mixed with the liquid stream exiting the washing unit, and some water is added to the settling tank to facilitate lignin precipitation. This mixture then enters a separator for the production of nearly pure lignin with about 10 % moisture (purity of 90 percent) [46–48]. Lignin is suitable for combustion due to its high heating value [29]. It can also be used for various applications, such as an organic material in agriculture and a carbon fiber source in industry [49]. The liquid stream is sent to the distillation column for ethanol recovery.

2.1.6. Ethanol recovery

The outlet stream from the top of the lignin separator is directed to a distillation column containing 26 trays, where the ethanol used in the pretreatment and washing units is recovered and returned to the pretreatment unit. The feed enters at tray 16, and the column operates at a pressure of 0.94 bar. In this unit, the ethanol concentration increases from 50 % to 93 %. About 80 % of the ethanol employed for pretreatment and washing is recovered and reused. To compensate for the remaining 20 %, the produced ethanol is utilized. In bioethanol production from olive waste, the recovery column bottom stream mainly contains about 90 % water, unconverted sugars, lignin, and yeast residues. Burning and disposal of these residues cause environmental problems, including the release of carbon dioxide [50]. To reduce the environmental impacts caused by traditional waste management methods, the liquid outlet stream from the bottom of the recovery column was mixed with the liquid stream from the solids and lignin separator and directed to the anaerobic digestion unit for biogas production, biofertilizer generation, which reduces containing COD up to 90 %.

2.1.7. Anaerobic digestion

The production of biogas includes two major steps: an anaerobic digestion unit and a biogas upgrading unit. The thermophilic dry anaerobic digestion (AD) method was employed in the previous process since it shortens both retention time and CAPEX. [31]. The anaerobic digester is an up-flow anaerobic sludge blanket (UASB) type reactor with a retention time of 25 days per feed at 37 °C [29]. The output stream from the anaerobic digestion is sent to a flash tank for the separation of liquid and gas phases. The gas stream (containing CH_4 and CO_2) is sent to the biogas upgrading unit. The bottom stream from the flash tank, which contains unconverted dissolved organic compounds, is separated as a biofertilizer (with 70 % moisture content) and sent for land application as fertilizer [51]. It is worth noting that one of our objectives in the present study was to minimize wastewater. All recyclable aqueous wastewater, such as the effluent from the flash drum, the second distillation tower, and the stripping tower, was recovered and reused. Other streams, such as the effluent from the bottom of the first tower and the wash liquid streams, were completely directed to the anaerobic digester to be used simultaneously with the olive waste for biogas and fertilizer production. A small amount of water is also used for washing and steaming (sterilization) of the fermenters during fermentation restart. However, due to its negligible quantity compared to the total water flow, it was disregarded.

2.1.8. Methane upgrading

The biogas stream exiting the flash tank is compressed to 10 bar in a compressor and cooled to 20 °C, then directed to the bottom of the absorption column, while water at 20 °C is fed from the top of the column. The upgraded biogas with a purity higher than 96 % exits from the top of the absorption column and, after being cooled to 1 °C, is sent to the biogas storage tank. The CO_2-rich water exits from the bottom of the absorption column and is transferred to the flash tank. To minimize methane loss, the pressure is reduced from 10 to 3 bar. The gas from the top of the flash column is returned to the biogas stream before the compressor. After exiting the flash column, the CO_2-rich solution was directed to the stripping column, where CO_2 is removed by a counter-current stream of air. The water is then returned to the top of the absorption column for reuse [51]. The gas leaving the stripping column contains CO_2 (over 90 %), along with small amounts of air and water, and is stored in a tank for use in the food industry [11]. The absorption and stripping columns were equipped with random packing, corresponding to 11 and 6 equilibrium stages [51], respectively. Table 3 reports the specifications of the scrubber, distillation, dehydration, ethanol recovery, absorber, and stripper columns for the set of A scenarios.

Table 3
Specifications of the scrubber, distillation, dehydration, ethanol recovery, absorber, and stripper columns for scenarios A-1 to A-5.

Scenario ID	Column specification	1st	2nd	3rd	4th	5th	6th
	No. of trays	7	22	30	22	11	6
	Feed plate column	1	2	15	2	11	1
A-1	Temperature, °C	37	60	80.44	60	20	20
	Column pressure, bar	1	0.97	0.9	0.94	10	1
	Type of condenser	–	partial	partial	total	–	–
	No. of trays	7	22	30	22	11	6
	Feed plate column	1	2	15	2	11	1
A-2	Temperature, °C	37	60	80.13	60	20	20
	Column pressure, bar	1	0.97	0.9	0.94	10	1
	Type of condenser	–	partial	partial	total	–	–
	No. of trays	7	22	30	22	11	6
	Feed plate column	1	2	15	2	11	2
A-3	Temperature, °C	37	60	80.4	60	20	20
	Column pressure, bar	1	0.97	0.9	0.94	10	1
	Type of condenser	–	partial	partial	total	–	–
	No. of trays	7	22	30	22	11	6
	Feed plate column	1	2	15	2	11	1
A-4	Temperature, °C	37	60	80.42	60	20	20
	Column pressure, bar	1	0.97	0.9	0.94	10	1
	Type of condenser	–	partial	partial	total	–	–
	No. of trays	7	22	30	22	11	6
	Feed plate column	1	2	15	2	11	1
A-5	Temperature, °C	37	60	80.33	60	20	20
	Column pressure, bar	1	0.97	0.9	0.94	10	1
	Type of condenser	–	partial	partial	total	–	–

Table 4
Specifications of the scrubber, distillation, dehydration, ethanol recovery, absorber, and stripper columns for scenarios B-1 to B-5.

Scenario ID	Column specification	1st	2nd	3rd	4th	5th	6th
	No. of trays	7	22	30	26	11	6
	Feed plate column	1	2	15	16	11	1
B-1	Temperature, °C	37	60	80.44	60	20	20
	Column pressure, bar	1	0.97	0.9	0.94	10	1
	Type of condenser	–	partial	partial	total	–	–
	No. of trays	7	22	30	26	11	6
	Feed plate column	1	2	15	16	11	1
B-2	Temperature, °C	37	60	80.13	60	20	20
	Column pressure, bar	1	0.97	0.9	0.94	10	1
	Type of condenser	–	partial	partial	total	–	–
	No. of trays	7	22	30	26	11	6
	Feed plate column	1	2	15	16	11	2
B-3	Temperature, °C	37	60	80.4	60	20	20
	Column pressure, bar	1	0.97	0.9	0.94	10	1
	Type of condenser	–	partial	partial	total	–	–
	No. of trays	7	22	30	26	11	6
	Feed plate column	1	2	15	16	11	1
B-4	Temperature, °C	37	60	80.42	60	20	20
	Column pressure, bar	1	0.97	0.9	0.94	10	1
	Type of condenser	–	partial	partial	total	–	–
	No. of trays	7	22	30	26	11	6
	Feed plate column	1	2	15	16	11	1
B-5	Temperature, °C	37	60	80.33	60	20	20
	Column pressure, bar	1	0.97	0.9	0.94	10	1
	Type of condenser	–	partial	partial	total	–	–

2.2. Set of B scenarios

In scenario set B, a similar feedstock supply as in scenario set of A is considered, which includes 15 ktons per year of olive stone (scenario B-1), 40 ktons per year of olive leaves (scenario B-2), 40 ktons per year of olive pomace (scenario B-3), 65 ktons per year of olive wood (scenario B-4), and 160 ktons per year of a mixture of all olive waste (scenario B-5), based on dry weight for the plant. In this scenario, a small amount of 1 % sulfuric acid is added as a catalyst to the pretreatment reactor. As a result, a significant portion of the lignin structure and lignocellulosic compounds are broken down, and sugar chains are converted into oligomeric compounds and transferred to the liquid phase [30]. All stages in this scenario set are similar to scenario set A, except for the sulfuric acid neutralization unit, which is placed after the ethanol recovery unit.

2.2.1. Sulfuric acid neutralization

The bottom stream of the ethanol recovery column contains not only water (90 %), unreacted sugars, and washed solids, but also sulfuric acid. The sulfuric acid is neutralized with ammonia, resulting in the production of ammonium sulfate. The produced ammonium sulfate is used as an agricultural fertilizer [33]. After acid separation, the remaining stream was sent to the anaerobic digestion unit, combined with the secondary stream from the solids and lignin separator, for biogas production. Other stages are similar to the scenario set of A. Table 4 shows the specifications of the scrubber, distillation, dehydration, ethanol recovery, absorber, and stripper columns for scenario set B.

2.3. Plant capacity and location

Since the primary olive production in Iran is concentrated in the northern regions, and due to the proximity of cities in this area, northern Iran has been identified as a suitable location for establishing the plant. Various plant capacities and different types of feedstocks have been taken into account in the planning process.

2.4. Process simulation and economics

Iran has a significant share in the production of oil and gas in the world. The burning of these fuels poses dire environmental challenges. Carbon emissions in Iran increased 2.5 times compared to the global annual average between 1990 and 2017, with the largest contribution coming from natural gas and oil [52]. Iran is known as one of the olive-producing countries, and according to statistics, the area under olive cultivation in this country has been reported to be over 38.5 thousand hectares [22]. Therefore, a techno-economic analysis of bioenergy production based on olive waste was conducted in this project in Iran. The Aspen Plus simulation software was utilized for the simulation of the main equipment. Additionally, Aspen Plus® v.11 was employed to solve material and energy balance equations and perform phase equilibria calculations using the equilibrium behavior is investigated by the non-random two-liquid (NRTL) thermodynamic model (liquid phase) and the Soave-Redlich-Kwong (SRK) equation (gas phase).

The physical and thermodynamic properties of olive waste were defined within the software based on the National Renewable Energy Laboratory (NREL) database for lignocellulosic components [53]. Economic assessments were then carried out based on the simulation

results, employing Aspen Process Economic Analyzer® (PEA) v.11. Sensitivity analysis was also conducted to identify the most influential variables in process economics. The prices of raw materials and products are presented in Table 5.

The plant was assumed to be located in Northern Iran, and the investment costs were adjusted using a location factor of 0.8. The target internal rate of return (IRR) was set at 20 %, while 15 % of the total project investment was allocated for contingencies. Working capital was estimated at 5 % of the total capital expenditure (CAPEX). Key economic parameters are summarized in Table 6.

3. Results and discussion

Based on experimental results [40], 10 scenarios for bioethanol-biomethane production from olive waste have been developed and optimized. Process simulations were performed using Aspen Plus, and their economic evaluations were conducted using Aspen PEA. The profitability and the impact of various parameters on the profitability of all scenarios were also assessed. Subsequently, the results will be presented and discussed.

3.1. Mass and energy balances

This study focuses on a biorefinery utilizing various olive wastes for bioethanol and biogas production, which is followed by an analysis of the results of the mass and energy balance. Tables 7 and 8 summarize the raw materials, products, and utility requirements for each scenario. Utilities include water, steam, and electricity. These two sets of scenarios include five distinct feedstock types (olive wood, pomace, stone, leaves, and a mixture of olive wastes) investigated with both ethanol (organosolv) and ethanol-sulfuric acid (acid-catalyzed organosolv) pretreatments.

The annual bioethanol production under scenarios A-1 to A-5 was 2.24, 4.18, 5.17, 10.16, and 22.50 ktons, respectively. For scenarios B-1 to B-5, the annual bioethanol production was 2.03, 5.17, 5.70, 9.50, and 23.44 ktons, respectively.

Based on the obtained results, it is observed that an increase in refinery capacity has led to a simultaneous increase in the amount of consumed materials and products. In the investigation of various scenarios, lignin separation using acid-catalyzed organosolv pretreatment has shown better performance compared to organosolv pretreatment, which could be due to improved pretreatment with acid. The lignin separation yield exhibited notable variation between the two investigated scenarios. In the set of A scenarios, the efficiency was found to range between 24–35 %, whereas in the set of B scenarios, the separation efficiency increased, ranging from 32-48 %. Sub-scenarios A-1 (35 %) and A-4 (34 %) in the set of A scenarios, and B-1 (48 %) and B-4 (40 %) in the set of B scenarios, demonstrated the highest yield of lignin separation. Based on the results, scenarios B-1 and subsequently A-1 exhibited the highest ratio of separated lignin to initial feedstock (139.5 and 101.32 L/kg, respectively) compared to other scenarios. On the other hand, scenario A-4 achieved the highest ratio of ethanol produced per feedstock capacity (198.5 L/kg). Additionally, scenarios B-2 and A-2 performed higher efficiency in biogas production. The ratios of products (ethanol, biogas, and lignin) to feed are shown in Fig. 2.

A gradual increase in CO_2 production with capacity is also expected. The plant fertilizer and lignin-rich stream were higher in scenario set B.

Examination of the results indicates that, at similar capacities, the use of acid-catalyzed organosolv pretreatment, compared to organosolv pretreatment, did not have a significant impact on the consumption of raw materials (such as enzymes) and the production of main and by-products. Furthermore, the consumption of utilities such as low-pressure steam and electricity was not significantly affected.

Table 5
The raw materials, products, and utilities with their price.

Items	Price ($/unit)	Reference
Raw materials		
Olive waste	17	a
Enzyme	1000	[30]
Sulfuric acid	262	[54]
Ammonia	300	[54]
Products		
Ethanol	1200	[11]
Biogas	1210	[31]
Lignin	630	[55]
CO_2	38	[11]
Rich-Lignin	42	[56]
Fertilizer	4.5	[57]
Utilities		
HP[b]	13.07	[32]
LP[c]	7	[32]
Water	0.06	[11]
Electricity	0.078	[11]

[a]Obtained from local farmers.
[b] HP: high-pressure steam (40 bar, 250 °C).
[c] LP: low-pressure steam (3 bar, 135 °C).

Table 6
The main economic parameters considered in the techno-economic analysis.

Parameter	Condition
Plant capacity	15 kton/year Olive stone waste /year (A1&B1) 40 kton/year Olive leaf waste /year (A2&B2) 40 kton/year Olive pomace waste /year (A3&B3) 65 kton/year Olive wood waste /year (A4&B4) 160 kton/year Mix all olive waste /year (A5&B5)
Plant Location	Asia, Iran, the CAPEX was calculated using a location factor of 0.8.
Year of analysis	2022
Depreciation	Straight line, zero salvage value, 7 years
Tax rate	0 %) According to Iranian tax law (
Operating hours	8000 h/year
Startup period	18 Weeks

3.2. Total investment cost

In the estimation of CAPEX, 49–73 % of fixed costs in different scenarios were allocated to equipment purchases. Given the emerging nature of biofuel production technologies and the existing uncertainties in this field, 15 % of total fixed costs have been allocated to this section to cover unforeseen expenses. Other costs, including piping and process equipment installation, electrical equipment installation, painting, contracts, and overhead, were determined as general and supervisory expenses. The total investment cost was estimated by Aspen PEA. Among the process units, the anaerobic digestion unit, due to the long residence time of anaerobic digesters for biogas production (25 days), the ethanol recovery section, due to the use of distillation columns, and the pretreatment section accounted for the largest portion of costs. Tables 9 and 10 show the CAPEX costs for all scenarios. Given that in these scenarios, all olive wastes were investigated separately with both pretreatments, and the waste capacities differ, the CAPEX will also differ. As the plant capacity increased, the production of bioethanol and biogas, and consequently the investment costs, also increased.

According to the economic results obtained for two scenarios, A-2 and B-2, with a feedstock capacity of 40 kton/year, scenario A-2, with a CAPEX of 41.85 M$ and a payout period of 8.19 years, is recognized as a more cost-effective option compared to scenario B-2, which has a cost of 48 M$ and a payout period of 10.97 years. Furthermore, both scenarios show economic improvement compared to the previous study [58], which reported a capacity of 30 kton/year and a total investment cost of 49.1 M$.

The economic results for two scenarios, A-4 and B-4, with a feedstock

Table 7
The annual raw materials, products, and utilities rates.

Scenario ID	A-1	A-2	A-3	A-4	A-5
Raw materials					
Olive waste (ton/year)	15,000	40,000	40,000	65,000	160,000
Enzyme (kg/year)	83	153	197	367	814
Products					
Ethanol (ton/year)	2244	4178	5170	10,162	22,490
Lignin (ton/year)	1521	2172	2709	4290	11,832
Biogas (m^3/year)	3,934,415	12,194,689	10,396,801	19,122,594	23,434,070
CO_2 (ton/year)	5179	20,387	13,559	27,462	67,119
Rich-Lignin (ton/year)	14,477	21,283	45,981	48,189	128,515
Fertilizer (ton/year)	44	32	32	53	108
Utilities					
Water (kg/year)	9,158,000	17,768,000	28,912,000	35,272,000	920,160,000
HP (ton/year)	103,493	273,678	277,043	44,808	1,105,017
LP (ton/year)	18,632,818	43,288,892	52,276,341	78,453,495	190,948,615
Electricity (kW)	2090	3222	3646	4473	7912

Table 8
The rate of raw materials, products, and utilities.

Scenario ID	B-1	B-2	B-3	B-4	B-5
Raw materials					
Olive waste (ton/year)	15,000	40,000	40,000	65,000	160,000
Enzyme (kg/year)	70	155	203	343	850
Sulfuric acid (ton/year)	137	368	368	602	1459
Ammonia (ton/year)	48	132	132	212	519
Products					
Ethanol (ton/year)	2028	4166	5696	9478	23,444
Lignin (ton/year)	2093	2489	3741	5265	14,197
Biogas (m^3/year)	3,912,361	12,119,149	10,347,642	19,118,558	22,460,957
CO_2 (ton/year)	5904	20,459	13,672	24,850	66,134
Rich-Lignin (ton/year)	9503	20,843	41,119	57,401	117,501
Fertilizer (ton/year)	650	1700	1703	2823	6871
Utilities					
Water (kg/year)	9158	7567	29,725	43,448	90,078
HP (ton/year)	103,552	273,837	277,202	448,341	1,105,647
LP (ton/year)	17,461,539	44,283,983	491,505,965	77,932,945	188,352,256
Electricity (kW)	1699	3927	3328	4422	7541

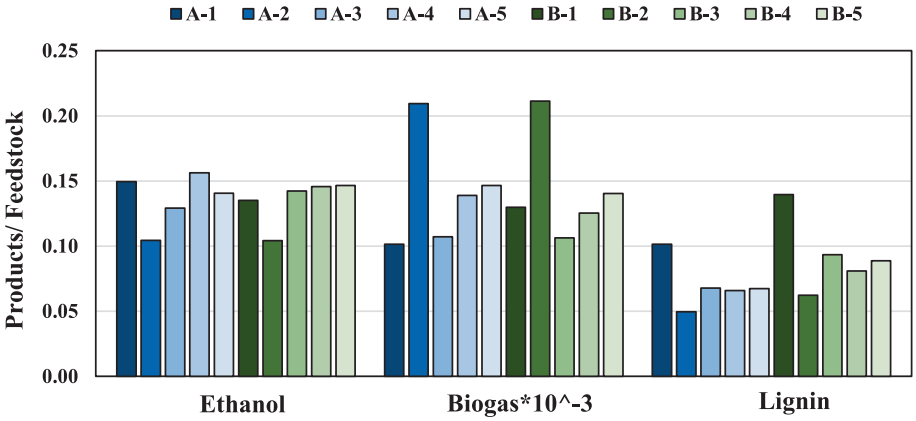


Fig. 2. The products-to-feedstock ratios for scenarios 1–5 A&B.

capacity of 65 kton/year, indicate that scenario A-4, with a CAPEX of 55.52 M$, and scenario B-4, with a cost of 55.42 M$, are significantly more cost-effective than the study by Solarte et al. [19]. In that study, a biorefinery with a capacity of 30 kton/year of olive tree pruning and a CAPEX of 24.52 M$ was designed for bioethanol production using dilute sulfuric acid pretreatment. This comparison indicates the higher economic efficiency of scenarios A-4 and B-4 at the industrial scale compared to previous studies.

Furthermore, the ratio of CAPEX to capacity in the study by Kautto et al. [59], which investigated a biorefinery with a capacity of 720 kton/

Table 9
Total project investment, equipment cost of units, and working capital of each A set scenario (M$).

Scenario ID	A-1	A-2	A-3	A-4	A-5
Feed Handling	0.4	0.55	0.55	0.6	1.23
Pretreatment	2.3	3.15	3.29	4.24	9.7
Hydrolysis &Fermentation	0.65	1.2	1.08	1.4	2.2
Distillation & dehydration	3.43	3.5	3.69	4	5
Lignin separation	0.8	1.03	1.18	1.5	4.17
Ethanol recovery	2.26	3.51	4.1	5.2	8.61
Anaerobic digestion	4.67	6.9	8.93	11.56	19.76
Methane upgrading	0.96	1.3	1.16	1.38	2.1
Utility	2.36	2.9	2.59	3	4.24
Working capital cost	1.67	2.1	2.3	2.87	3.8
CAPEX	33.4	41.85	46	55.52	75.74

Table 10
Total project investment, equipment cost of units, and working capital of each B set scenario (M$).

Scenario	B-1	B-2	B-3	B-4	B-5
Feed Handling	0.4	0.55	0.55	0.6	1.23
Pretreatment	2.3	3.15	3	4.21	9.77
Hydrolysis &Fermentation	0.61	0.83	1.34	1.5	2.2
Distillation & dehydration	3.37	3.51	3.6	4	4.8
Lignin separation	0.759	1.2	1.12	1.5	4.2
Sulfuric acid neutralization	0.42	1.02	0.86	1.14	2.48
Ethanol recovery	2.31	3.52	4.12	5.2	8.6
Anaerobic digestion	3.6	9.68	7.93	10.9	17.5
Methane upgrading	0.98	1.3	1.12	1.38	1.8
Utility	2.4	2.84	2.53	3	4.6
Working capital cost	1.6	2.41	2.29	2.8	3.85
CAPEX	31.96	48.01	45.83	55.42	77

year of hardwood logs and a CAPEX of 846 M$ using organosolv pretreatment for bioethanol production, is comparable to the results of scenarios A-4 and B-4. This highlights the scalability and economic stability of these two scenarios.

Regarding the two scenarios A-5 and B-5, with an annual feedstock capacity of 160 kton, economic results indicate that scenario A-5, with a CAPEX of 75.74 M$, and scenario B-5, with a CAPEX of 77 M$, are significantly more cost-effective than the study by Servian et al. [60]. In this study, a biorefinery with a capacity of 36 kton/year of olive tree pruning and a CAPEX of 39.5 M$ was investigated for bioethanol production using steam explosion pretreatment. Additionally, in the study by Kabir et al. [61], a biorefinery with a capacity of 20 kton/year of native forest residues and a CAPEX of 59.4 M$, utilizing organosolv pretreatment for biogas production, was examined. Comparisons with scenarios A-5 and B-5 indicate that, at a larger scale, these two scenarios possess a higher economic advantage compared to similar projects at smaller scales. These comparisons suggest that the more scalable scenarios in this research hold a significant competitive economic advantage, demonstrating the high potential of olive waste for industrial-scale biofuel production.

Based on the obtained results, the cost of the anaerobic digestion process in Group A scenarios was generally higher than in Group B scenarios. However, scenario B-2, exceptionally, showed a higher cost compared to scenario A-2, which necessitates more detailed investigations and supplementary analyses.

On the other hand, with an increase in plant capacity, costs associated with ethanol recovery and biomethane purification also rise. However, this increase is mitigated by economies of scale, resulting in a proportionally lower increase compared to the growth in capacity. For instance, in scenario A-1, the ethanol recovery cost is reported as 2.26 M $, whereas for scenario A-5, with a capacity more than tenfold greater, this figure is approximately 8.61 M$.

From an overall review of the results, it can be inferred that the CAPEX increases with both higher plant capacity and increased complexity of the input raw material composition. When comparing the scenario groups, it's observed that Group B scenarios generally exhibit lower CAPEX. A notable point, based on this economic analysis, is that the use of an acid catalyst in the pretreatment stage does not create a significant advantage in reducing costs. The addition of an acid as a catalyst during the pretreatment stage enables more efficient lignin separation, leading to significant improvements in process efficiency [37]. This approach permits the use of smaller equipment in subsequent stages, thereby reducing both CAPEX and OPEX. However, the requirement for acid neutralization following pretreatment increases overall expenses.

3.3. Cost distribution

The distribution of production costs for all scenarios is illustrated in Fig. 3. Production costs are categorized into three main sections: direct production costs, fixed production costs, and general production costs. Direct production costs encompass expenses related to raw materials, labor, supervision, utilities, maintenance, and repairs. Fixed production costs encompass taxes, insurance, depreciation, and plant overhead. General costs include administrative expenses, research and development (R&D), and finance [62]. Based on the results, the largest portion of expenses is attributed to raw material procurement, with olive waste accounting for the majority of these feedstock costs. The co-products generated in these scenarios include biogas, pure lignin, lignin-rich product, CO_2, and biofertilizer.

3.4. Profitability assessment of the scenarios

Tables 11 and 12 present the economic evaluations of the scenarios based on key metrics, including the profitability index (PI), net return rate (NRR), and payout period (PP). The NRR is an indicator of profitability, calculated as the net present value divided by the sum of the present value of costs, expressed as a percentage. This index is positive for all scenarios except A-1 and B-1, indicating that the total net present value is higher than the present value of costs. A negative value for this index signifies a negative net present value.

Comparing these parameters, it's clear that all scenarios demonstrate greater economic efficiency compared to olive stone waste. This is primarily due to the higher capacity of other waste types relative to olive stone (15 kton/year).

The payout period is defined as the time required to recover the total initial investment. According to the results presented in the relevant tables, scenarios A-1 and B-1, despite having low initial CAPEX, are deemed economically unfeasible due to their low plant capacity.

Despite their economic viability, scenarios A-2 to A-4 and B-2 to B-4 exhibit long payout periods due to the low capacity of olive wood, leaf, and pomace waste. By blending these wastes with agricultural residues (possessing similar structural compositions) or by mixing all olive wastes, it is possible to increase their production capacity and economic efficiency, thereby reducing the payout period.

The profitability index (PI) indicates the relative profitability value compared to the costs. If this index is greater than one, it signifies that the total net present value is higher than the present value of the costs, and the project can be economically viable. Conversely, if it is less than one, it indicates a negative net present value, meaning the project will be unprofitable.

This criterion was greater than one for all conducted scenarios except A-1 and B-1. Among the scenarios, B-5 with a profitability index of 1.78 (in the B set) and A-5 with a profitability index of 1.75 (in the A set) are economically more favorable than the other scenarios. Furthermore, the results indicate that the economic indicators for scenario set A were generally superior to those in scenario set B. Plant life is intended to be 20 years, and the payout period varies depending on the scenario.

In Fig. 4, the yearly profit and CAPEX for each scenario are presented. The annual profit was calculated using eq. (1):

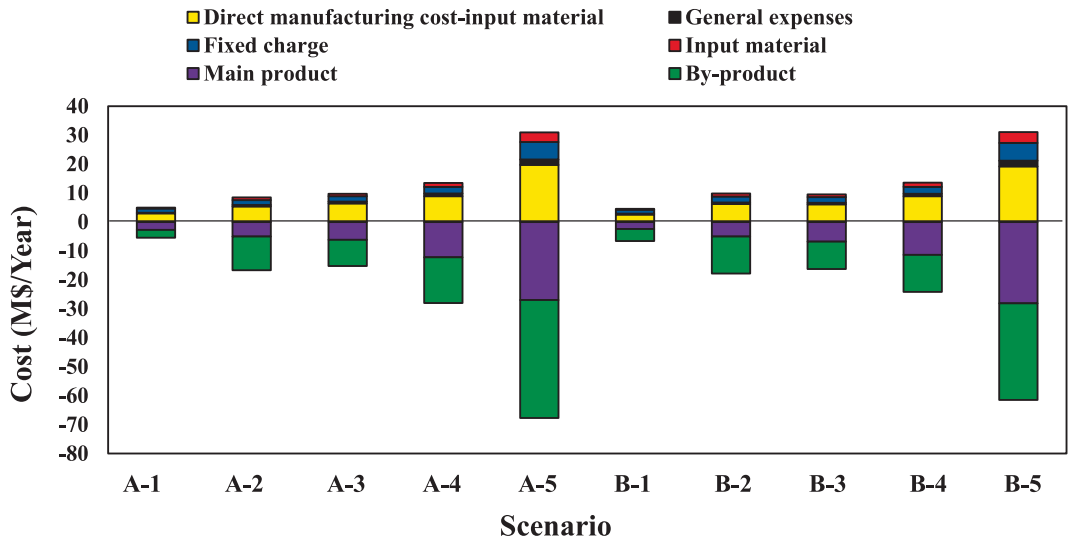


Fig. 3. Breakdown of the Operational expenditures (OPEX) for all scenarios.

Table 11
The profitability parameters for scenario set A.

Scenario ID	Payout period (year)	Net return rate (NRR)	Profitability index (PI)
A-1	0	−29.31	0.7
A-2	8.19	32.56	1.32
A-3	18.08	2.13	1.02
A-4	6.96	43.15	1.43
A-5	4.28	76.42	1.75

Table 12
The profitability parameters for scenario set B.

Scenario ID	Payout period (year)	Net return rate (NRR)	Profitability index (PI)
B-1	0	−19.98	0.8
B-2	10.66	18.79	1.17
B-3	12.84	11.07	1.11
B-4	7.42	37.98	1.38
B-5	4.23	79.05	1.78

$$YP = \left(\frac{PSC - \mathrm{OPEX}}{CAPEX}\right) * 100 \quad (1)$$

Where YP is the yearly profit, OPEX is the operational expenditures, CAPEX is the total capital investment costs, and PSC is the product selling costs. In this equation, OPEX has been reduced from PSC, and the amount obtained on CAPEX was divided.

Based on the results obtained, in both scenario sets of A and B, increasing the processing capacity leads to higher profitability and CAPEX. As shown in Fig. 6, scenario A-1 with a capacity of 15 kilotons per year yields an annual profit of 5.3 % and requires a total CAPEX of 33.4 M$. In contrast, scenario A-5 with a capacity of 160 ktons per year achieves an annual profit of 55.72 % and incurs a CAPEX of 75.74 M$. Olive leaves and wood (scenarios A-2, A-4, B-2, B-4) also demonstrate favorable profitability, which may be attributed to their high content of fermentable sugars.

The results indicated that using olive pomace and stone as feedstock, combined with acid-catalyzed organosolv pretreatment, results in better economic performance (higher annual profit and lower CAPEX) compared to organosolv pretreatment. This improvement can be attributed to more effective lignin removal during the pretreatment stage, resulting in higher hydrolysis and fermentation yields, which ultimately lead to greater production of bioethanol and biogas.

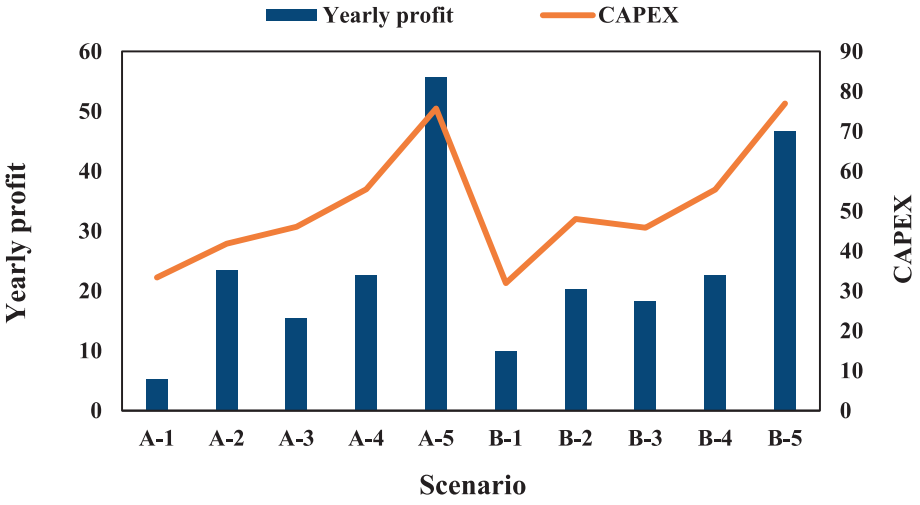


Fig. 4. Yearly profit (%) and CAPEX (M$) for all scenarios.

However, compared to other scenarios, olive pomace and stone, particularly scenarios A-1 and B-1 with lower capacities, have shown lower profitability. In contrast, olive leaves (scenario A-2, with an annual profit of 23.44 %) have demonstrated better performance in comparison with olive wood residues, likely due to their higher glucan content. In the organosolv pretreatment process, this scenario achieved higher annual profit and lower CAPEX compared to the acid-catalyzed organosolv method.

As can be inferred from the results, profitability is directly related to plant capacity. However, process efficiency depends on the type of pretreatment and the composition of the feedstock. Scenario A-5, with a CAPEX of 75.74 M$, shows only a slight difference compared to B-5, which requires 77 M$ (due to costs associated with acid consumption and neutralization). Nevertheless, A-5 yields a higher annual profit.

This may be due to higher efficiency in the hydrolysis and fermentation processes, resulting in greater product yields. The highest annual profit in these two scenarios could be attributed to the integrated utilization of all olive waste materials. This indicates the economic advantage of comprehensively exploiting all waste streams of the olive industry. Moreover, since the chemical composition of individual lignocellulosic olive wastes differs, the optimal pretreatment conditions for each fraction are likely to vary. In fact, several studies have reported preprocessing optimization strategies for different types of olive waste [19,38,63]. A single set of conditions previously reported and validated [40] was applied in this study. The overall process economics could potentially be improved in the future by applying fraction-specific pretreatment conditions once sufficient experimental data become available.

Although the present study uses Iran as a representative case study, the developed framework and modeling approach are universally applicable. The methodology can be readily adapted to other regions by adjusting location-specific parameters such as energy costs, labor rates, and taxation structures, since these parameters depend on local conditions and regulations [31,45,55,64]. This regional case study therefore, serves as a practical example demonstrating how the proposed framework can be implemented globally under varying economic and technical conditions.

3.5. Sensitivity analysis

Changes in the cost of raw materials, water, and electricity, by-product prices, plant location, plant capacity, and equipment size can affect the economic output parameters (such as profitability indicators). For this reason, a sensitivity analysis has been conducted to identify the impact of variations in product price, by-product price, electricity cost, and raw material cost on the profitability indicators of the processes. To

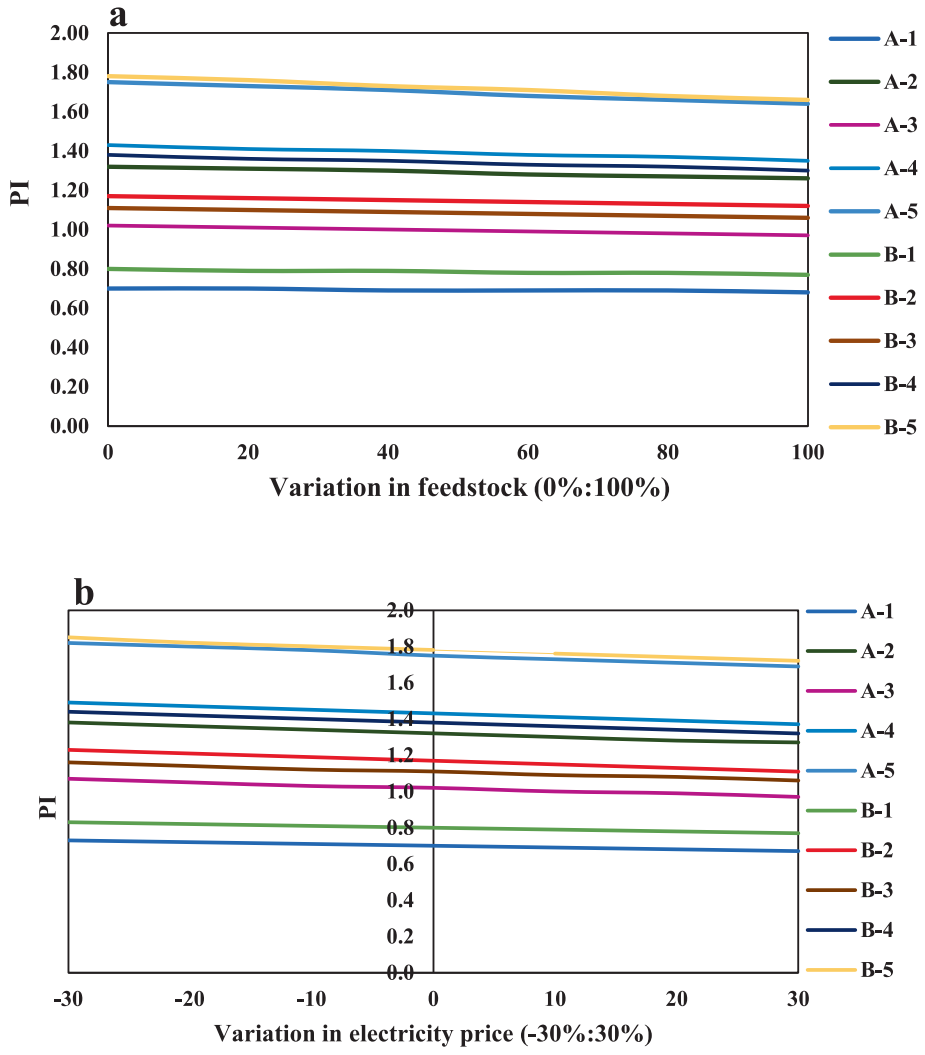


Fig. 5. Effect of (a) feedstock and (b) electricity price variation on the PI (Profitability Index).

assess the impact of plant capacity on total investment and profitability indicators across all scenarios, an increase in feedstock capacity (on a dry basis) from 100 − 600 % was evaluated.

In Fig. 5, the impact of changes in electricity price (ranging from −30 to + 30 %) and raw material cost (olive waste) (ranging from 0 to 100 %) on the profitability index (PI) for all scenarios is analyzed. The findings have shown that a rise in the price of electricity and raw materials causes the PI to fall in every scenario. According to the findings, a 100 % increase in the price of olive waste caused a decrease of 0.1–0.2 units in the Profitability Index (PI). In contrast, a 30 % increase in the price of electricity led to a more significant decrease in the Profitability Index (between 0.13 and 0.16 units), indicating a stronger impact of this factor on profitability.

Scenarios A-5 and B-5 have shown the most consistent performance out of all the scenarios that were evaluated. These scenarios have demonstrated their economic viability even in the face of extremely

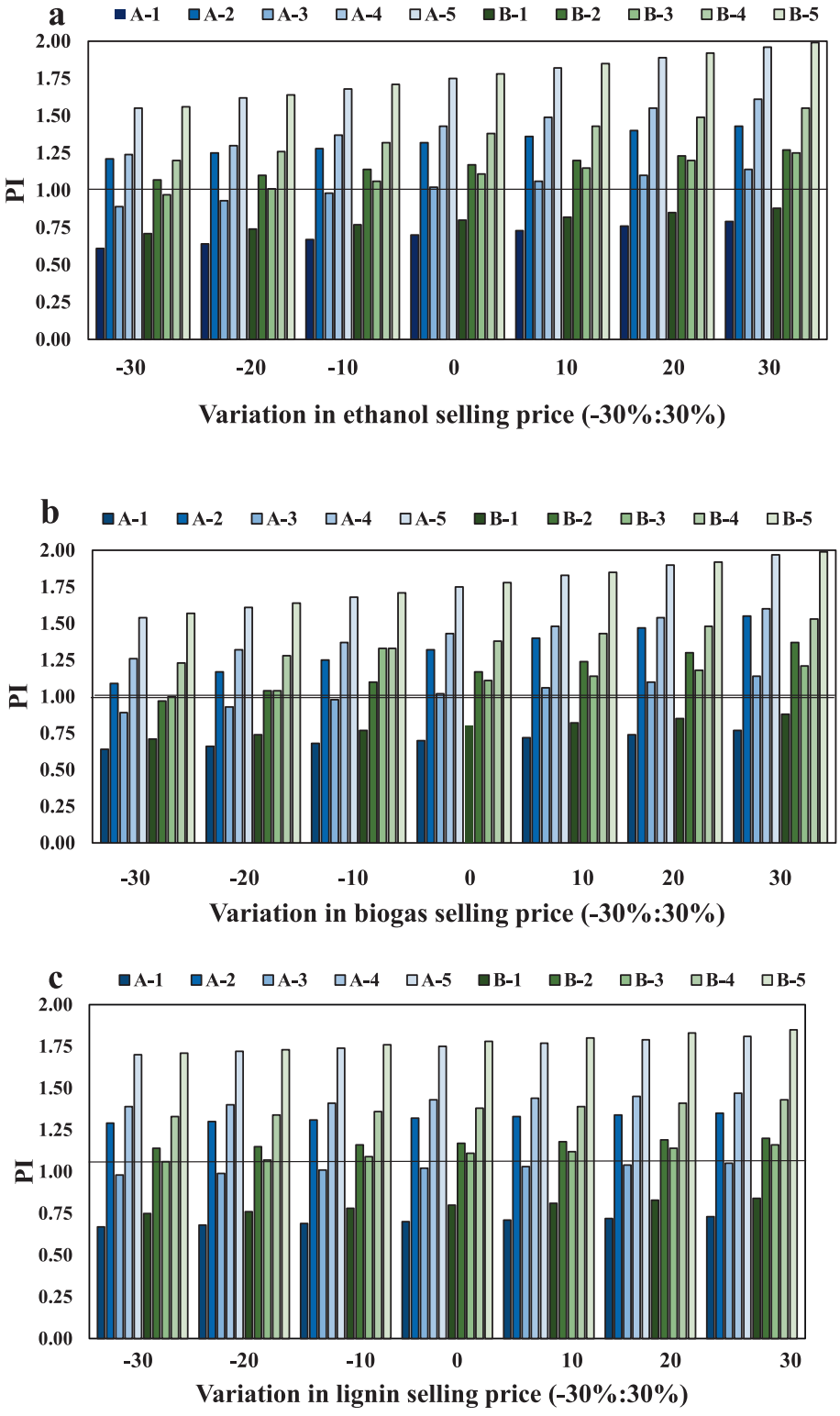


Fig. 6. Effect of (a) product and (b and c) by-product selling price variation on the PI (Profitability Index).

pessimistic circumstances by managing to maintain a PI > 1.5 even when raw material and electricity prices have increased simultaneously. They are now viable options for industrial-scale deployment due to their resilience. In contrast, sensitivity analysis to examine the impact of plant capacity on the profitability of the scenarios has shown that a doubling of scale increases the performance index of these two scenarios (A-1 and B-1) by 1.03 and 1.16, respectively.

The analysis's findings indicate that, when it comes to increasing project profitability, energy cost management is more important than raw material price control. Even though the mid-range scenarios (A-2 to A-4 and B-2 to B-4) have been economically viable under normal conditions, a significant rise in the price of raw materials could push them over the edge of viability.

Examining the changes in the Profitability Index (PI) in Fig. 6 has shown that price fluctuations (−30 to + 30 %) of the final products (ethanol, biogas, and lignin) had a direct and significant impact on the profitability of the project. Even with a 30 % decrease in ethanol price, scenarios A-5 and B-5 have the highest PI values (above 1.5), which confirms their high stability against ethanol price variations. On the other hand, for A-1 and B-1 scenarios, the smallest PI is observed, and the lowest sensitivity to ethanol price changes is shown. Among the various products, ethanol had the greatest impact on the profitability index. Scenarios A-5 and B-5 were able to maintain PI above 1.5 even with a 30 % decrease in ethanol price, indicating the high resilience of these scenarios to market fluctuations. In contrast, scenarios A-1 and B-1 reached PI of about 0.3 with a 30 % decrease in ethanol price, indicating the extreme vulnerability of these scenarios and their lack of economic justification in such conditions.

Biogas also had a considerable impact on the profitability of the project, although the intensity of this impact was less than that of ethanol, which indicates a lower share of biogas revenue in total revenues compared to ethanol. Lignin had the least impact on the profitability index. This product has a smaller share of total revenue compared to ethanol and biogas in terms of both price and production volume. For this reason, lignin price fluctuations have had a more limited impact on PI. The study highlights the significance of diversifying products and increasing high-value ones like ethanol, suggesting scenarios A-5 and B-5 as optimal investment options for industrial scale.

The minimum economically viable capacity (C_{min}), also known as the break-even point, is defined as the smallest production capacity at which a process or plant becomes economically feasible, corresponding to the point where the PI equals one. The calculated C_{min} values for each scenario are presented in Table 13. The results indicate that refineries operating below these capacity thresholds are not economically viable. The lowest minimum capacities were observed in sub-scenarios A-1 and B-1, while the highest values were found in sub-scenarios A-5 and B-5. This suggests that the minimum economic capacity of the refinery increases as a result of mixing different types of olive waste.

This finding suggests that the high initial investment cost represents a major limitation for small olive oil producers. For such producers, alternative processing strategies, e.g., batch fermentation and batch downstream processing, may offer more practical solutions compared to the continuous systems considered in this study. Although batch processes may yield lower overall productivity, their reduced capital requirements and operational flexibility could make them more economically attractive for small-scale applications. Future studies could explore the techno-economic feasibility of these alternative configurations to identify optimal solutions for producers of different scales.

Table 13
The minimum economically viable capacity for each scenario.

Scenario	A-1	A-2	A-3	A-4	A-5	B-1	B-2	B-3	B-4	B-5
C_{min} (kton/year)	30	34	36	49	80	30	32	40	45.5	88

3.6. Comparison of scenarios with other studies

In this study, a process for bioethanol production from all types of olive waste was proposed, based on organosolv and acid-catalyzed organosolv pretreatments, followed by enzymatic hydrolysis and PSSF. Ten scenarios were defined and compared. To date, several techno-economic studies have examined various feedstocks and combinations of pretreatment, hydrolysis, fermentation, and purification methods. An overview of the comparison of the gasoline equivalent price of some of these processes is provided in Table 14.

For a more accurate and meaningful comparison of the produced fuel products, their gasoline-equivalent prices were calculated [29]. These analyses were conducted on a fuel energy content basis of the products. The calculations assumed the following lower heating values (LHV): ethanol (21.2 MJ/L), methane (36.1 MJ/Nm^3), and gasoline (32 MJ/L). Note that the LHV is defined by assuming the water produced during combustion is in the vapor phase. The gasoline-equivalent price for every fuel is the amount that fuel would cost to provide the same amount of energy as one liter of gasoline. Therefore, the energy of ethanol and biogas produced out of 15, 40, 40, 65, 160 ktons/year of stone, leaves, pomace, wood, and a mixture of all waste are equivalent to 3.6, 12.96, 9.18, 18.7, 45.32 million liters of gasoline, respectively (scenarios set of A). The energy of the ethanol and biogas produced out of a 160 ktons/year mixture of all waste is equivalent to 45.32 million liters of gasoline, which is consistent with the results of Shafiei et al. [45]. Additionally, the energy of the ethanol and biogas produced in scenario set B is in the range of 3.9—45 million liters. The gasoline-equivalent price for the scenarios sets A and B is in the range of 0.55——1.03 and 0.57——0.87 $/L, respectively (Table 14).

The economic results indicate that scenarios A-5 and B-5 provide the highest amount of gasoline-equivalent (45320 and 45,000 cubic meters) with the lowest gasoline-equivalent price (0.56 and 0.57 $/L), which indicates high energy efficiency in the use of mixed olive waste. In contrast, the least amount of energy is generated in scenarios A-1 and B-1 (3600 and 3900 cubic meters of gasoline-equivalent). They also have the lowest economic efficiency and comparatively high gasoline equivalent costs (1.03 and 0.87 $/L), which may be related to olive stone's limited capacity.

In the scenario set A, the average price of energy produced is lower in most scenarios than in scenarios set B, except for A-1, which has a higher cost. On the other hand, in scenario set B, some energy efficiency gains are observed, but they come at the cost of additional costs due to the use of acid and the need for neutralization. For example, scenario B-2 (olive leaves), with 13,000 cubic meters of gasoline-equivalent produced for 0.62 $/L, is only slightly more efficient but more expensive than scenario A-2 (12960 cubic meters and 0.55 $/L). This comparison is also true for other scenarios (e.g., B-4 vs. A-4, or B-3 vs. A-3), indicating that the technical advantage of using acid pretreatment at this scale may not be justified by its operational costs.

In comparison, the study by Shafiee et al. [29], which investigated biogas production from wheat straw using steam explosion pretreatment, has a gasoline-equivalent cost of 0.56–0.67 $/L, which is in the same range as or higher than the olive scenarios. Also, biogas from waste paper tubes [29] and ABE from sweet sorghum [32] have a higher cost than most of the A and B scenarios, with a price of 0.68–0.79 and 0.68 $/L. While biodiesel-biogas from castor and Eruca sativa [31], with much lower prices (0.32 and 0.37 $/L), have high economic performance, the need for arable land limits them from a long-term sustainability perspective.

However, the cost of producing biogas from native forest residues [61] treated with organosolv is high (1.81 $/L), which reflects the possible logistical difficulties involved in the exploitation of forest resources. In addition to being expensive (0.84–0.90 $) and directly competing with the food chain, biogas derived from wheat and sugar beet [65] is dubious from a sustainability standpoint.

According to the results, using olive waste (particularly in scenarios

Table 14
Comparison of scenarios with other techno-economic analyses.

Raw material	Pretreatment	Product	Gasoline equivalent (ML)	Gasoline equivalent price ($/L)	Reference
Olive stone	Organosolv	Bioethanol – biogas	3.6	1.03	This study A-1
Olive leaf	Organosolv	Bioethanol – biogas	12.96	0.55	This study A-2
Olive pomace	Organosolv	Bioethanol – biogas	9.18	0.88	This study A-3
Olive wood	Organosolv	Bioethanol – biogas	18.7	0.61	This study A-4
Mixture of all waste	Organosolv	Bioethanol – biogas	45.32	0.56	This study A-5
Olive stone	Acid-catalyzed organosolv	Bioethanol – biogas	3.9	0.87	This study B-1
Olive leaf	Acid-catalyzed organosolv	Bioethanol – biogas	13	0.62	This study B-2
Olive pomace	Acid-catalyzed organosolv	Bioethanol – biogas	9.6	0.84	This study B-3
Olive wood	Acid-catalyzed organosolv	Bioethanol – biogas	17.2	0.68	This study B-4
Mixture of all waste	Acid-catalyzed organosolv	Bioethanol – biogas	45	0.57	This study B-5
Wheat straw	Steam explosion	Biogas	–	0.56–0.67	[29]
Paper tube	Steam explosion	Biogas	–	0.68–0.79	[29]
Castor	MeOH-H_2SO_4	Biodiesel – biogas	–	0.32	[31]
Eruca sativa	MeOH-H_2SO_4	Biodiesel – biogas	–	0.37	[31]
Sweet sorghum	Acetone – H_2SO_4	ABE	–	0.68	[32]
Native forest residues	Organosolv	Biogas	–	1.81	[61]
Wheat- sugar beet	Yes	Biogas	–	0.9	[65]
Wheat	Yes	Biogas	–	0.84	[65]

A-5 and B-5) is a wise way to produce renewable energy because it is readily available, inexpensive, and has a high potential for energy conversion when compared to other sources. Large-scale sustainability will only be achievable if resources are used without food competition and the need for agricultural land (such as olive waste), even though some alternative methods might seem less costly.

4. Conclusions

This research aims to determine the technological and economic feasibility of producing bioethanol, biogas, and lignin from olive waste. The results indicate that ethanol production from lignocellulosic olive waste is economically viable using organic solvent pretreatment and organic solvent pretreatment with acid catalyst, as well as PSSF. Olive waste mixture as a suitable feedstock is a suitable option for the development of biorefineries. The techno-economic analysis also shows that bioethanol production from these wastes will be economical if biogas, pure lignin, and biofertilizer are also obtained as by-products from the plant residues. Among the scenarios studied, the two scenarios A-5 (organosolv pretreatment) and B-5 (acid-catalyzed organosolv pretreatment) had the best economic indicators. However, the use of acid catalyst did not have a significant effect on annual profit, total investment cost, and profitability indicators.

The gasoline-equivalent cost varies between 0.55 and 1.03 for the plant with a capacity of 15–160 ktons/year of olive waste. Sensitivity analysis also highlighted the importance of policy interventions to optimize energy consumption and reduce electricity costs, as fluctuations in electricity prices directly and significantly impact the project's profitability index. It would be valuable to compare the results of the present study with those of a similar process involving biodiesel production from olive pomace oil, including life cycle assessment and process exergy analysis.

CRediT authorship contribution statement

Maryam Jafari-Gohargan: Writing – original draft, Visualization, Software, Resources, Investigation, Formal analysis, Data curation. **Mohsen Mohammadi:** Writing – review & editing, Supervision, Methodology. **Arjomand Mehrabani-Zeinabad:** Writing – review & editing, Supervision, Methodology. **Keikhosro Karimi:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fuel.2025.137530.

Data availability

No data was used for the research described in the article.

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